CONCEPT NOTE

Forecasting Scope & Modelling Approach

Procurement Demand Forecasting Initiative

EHA Clinics · Supply Chain Analytics

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This document defines the scope boundary for the procurement demand forecasting model, including commodity coverage, facility selection, time horizon, forecast granularity, and the modelling approach to be applied.

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1. Purpose

This concept note defines the boundaries within which the procurement demand forecasting model will operate. It answers four foundational questions: *what* will be forecast, *where*, *over what time period*, and *at what level of detail*.

Establishing a clear scope before modelling begins prevents scope creep, focuses analytical resources on the data most likely to yield accurate and actionable forecasts, and provides an agreed reference point for evaluating whether the KPI target — a Mean Absolute Percentage Error (MAPE) of $\leq 20\%$ for critical health commodities by Q3 2026 — has been met.

2. Background and Context

KPI target (as stated by management):
"Establish current forecast accuracy baseline using historical Odoo/LoMIS data by Q2, then achieve MAPE $\leq$ 20% for critical health commodities across pilot facilities by Q3."

The primary dataset is the EHA Clinics procurement record extracted from the Odoo ERP system. It spans **2018 to 2024**, covers **16 branch locations** across Lagos, Abuja, and Kano, and contains more than 50 distinct procurement categories — from prescription medicines to vehicle repairs and staff training.

Because procurement transactions encode both clinical consumption (medicines, consumables) and institutional overhead (office supplies, maintenance), the raw dataset cannot be modelled as-is. A well-defined scope is necessary to isolate the signal that corresponds to patient-driven demand.

3. Commodity Scope — What to Forecast

The forecasting model will target **critical health commodities**: items directly consumed in patient care whose demand is driven by consultation volumes and recurring clinical need. The categories below are identified from the dataset's `category_name` field and are confirmed in-scope for the pilot.

Category	Rationale
Prescription Medications	Core clinical demand driver; high-volume, recurring
Over The Counter Drugs	High throughput; strong seasonal signal
Vaccines / NPI Vaccines	Programme-driven with seasonal and calendar patterns
Injections	Procedure-linked consumable; forecastable against patient-visit data
Medical Consumables	Direct patient-care materials (cannulas, gloves, etc.)
Laboratory Consumables	Tied to diagnostics throughput; recurring
Dental Consumables	Specialty but volume-trackable across pilot sites

3.1 Explicitly Excluded Categories

The following categories are **out of scope** for forecasting. They are either non-recurring, driven by administrative or capital decisions rather than patient demand, or produce so few transactions that modelling is not meaningful:

- Office Operations, General Assets & Expenses, Office Expense and Supplies
- Vehicle, Facility & Repairs, Furniture and Fittings
- Specialist Services, Outsourced Services, Deliveries
- Travel, Rent & Leases, Events, Research
- Services – Asset Purchase, Laboratory Instruments, Medical Equipment (capital items)
- `xxx ARCHIVED xxx` (deprecated catalogue entries)

3.2 Order Status Filter

Only records where `state = done` are used. These represent confirmed, delivered purchase orders that reflect true realised demand. Records with `state = purchase` (orders confirmed but not yet receipted) are excluded from training data to avoid inflating demand estimates.

4. Facility Scope — Where to Forecast

The dataset contains 16 named branches. For the pilot phase, forecasting will focus on the six facilities with the longest continuous procurement history and sufficient monthly transaction volume to support model training.

Facility	City	Notes
Abuja – Asba and Dantata	Abuja	Largest transaction volume in the dataset
Abuja – Lugbe	Abuja	Consistent ordering history
Kano – Lamido Crescent	Kano	High-volume; strong seasonal patterns expected
Kano – Independence Road	Kano	Representative urban clinic
Lagos – Sangotedo Ajah	Lagos	Only Lagos facility with sufficient history
REACH Abuja Hub 1 – Kuje	Abuja	Community-facing; captures different demand profile

> **Note on Kano Distribution Warehouse:** This branch will be treated as a *supply-side reference*, not a demand point. Its procurement records reflect centralised bulk stocking decisions and do not represent direct facility consumption. It may be used later to model replenishment lead times.

5. Time Horizon

5.1 Training Window

January 2021 – December 2023 (36 months).

While the dataset begins in 2018, records from 2018–2020 span the clinic expansion period and the COVID-19 disruption. Including these years without explicit adjustment would embed abnormal demand patterns — procurement surges, facility closures, and supply chain disruptions — into the training data. The post-2020 period reflects more stable, representative clinical operations and is therefore the primary training window.

5.2 Validation Window

January 2024 – June 2024 (6 months).

This held-out period is used to compute the Q2 MAPE baseline. Comparing model predictions against actuals in this window gives an objective, pre-modelling measure of current forecast accuracy.

5.3 Forecast Horizon

Monthly, rolling **3-month forward** window, extendable to 6 months once model performance is validated. Monthly granularity aligns with the existing `month` column in the dataset and mirrors typical procurement cycle length at these facilities.

6. Forecast Granularity

The fundamental unit of forecast is:

$$\underbrace{\text{Commodity Category}}_{\text{what}} \times \underbrace{\text{Facility}}_{\text{where}} \times \underbrace{\text{Month}}_{\text{when}}$$

Individual SKU-level forecasting (e.g., *Amoxicillin 500mg × 100 at Asba – monthly*) is a natural next step but will not be attempted in the pilot phase. SKU-level records are currently too sparse at several facilities to produce statistically stable models, and the measurement overhead would complicate the Q2 baseline assessment. SKU-level modelling will be scoped separately once category-level MAPE $\leq 20\%$ is achieved.

7. Primary Performance Metric

Forecast accuracy will be measured using the **Mean Absolute Percentage Error (MAPE)**:

$$\text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{A_t - F_t}{A_t}\right| \times 100$$

where A_t is actual procurement quantity in period t and F_t is the forecast quantity. MAPE is the standard indicator for supply chain forecast accuracy and is directly aligned with the stated KPI.

MAPE will be computed **per category-facility pair** to identify which commodities and locations are under- or over-performing, enabling targeted model improvements ahead of the Q3 deadline.

8. Modelling Approach

8.1 Guiding Principle

The modelling strategy follows a **tiered complexity** approach: begin with the simplest model that is capable of meeting the MAPE $\leq 20\%$ target, and introduce additional complexity only when simpler models demonstrably fall short. This principle keeps the pipeline interpretable for supply chain staff, reduces the risk of overfitting on a 36-month training window, and ensures that model outputs can be explained to — and trusted by — facility managers.

> **Core rule:** A more complex model is justified only if it reduces validation MAPE meaningfully (by at least 3–5 percentage points) compared to the tier below it. Complexity for its own sake will not be pursued.

8.2 Tier 1 — Benchmark Models (Baseline)

Benchmark models are applied first and serve two purposes: they establish the Q2 MAPE baseline required by the KPI, and they set the performance floor that all subsequent models must beat to justify their use.

- **Seasonal Naïve:** The forecast for month t equals the actual value from the same month in the prior year ($\hat{F}_t = A_{t-12}$). This is the most widely used operational baseline in health supply chain contexts.
- **12-Month Rolling Average:** The forecast equals the mean of the preceding 12 months' actuals. Smooths short-term noise without requiring any model fitting.

If either benchmark already achieves MAPE $\leq 20\%$ for a given category-facility pair, that

pair is considered solved at the baseline stage and is not escalated to higher tiers.

8.3 Tier 2 — Classical Time Series Models

Classical statistical models are the primary candidates for the pilot phase. They are well-suited to monthly health commodity data, require relatively little data volume, and produce uncertainty intervals that are operationally meaningful for safety-stock planning.

Method	Description and Rationale	Best suited for
ETS (Error-Trend-Seasonality, Holt-Winters)	Decomposes each series into error, trend, and seasonal components. Parameters are fitted by maximum likelihood; the model automatically selects additive or multiplicative seasonality. Well-established for monthly pharmaceutical and consumable demand.	Series with clear trend and/or seasonality; moderate data volume
SARIMA (Seasonal ARIMA)	Models autocorrelation structures in the series alongside seasonal differencing. Captures carry-over demand effects (e.g., a stock-out in one month suppressing observed procurement in the next). Order selection via AIC/BIC grid search.	Series with autocorrelation; irregular ordering patterns
Theta Method	Decomposes the series into two "theta lines" (long-run trend and short-run variation) and recombines them. Computationally cheap and consistently competitive in empirical benchmarks (M3 and M4 competitions).	Sparse or noisy series; rapid iteration

All Tier 2 models will be fitted independently per category-facility pair, reflecting the expectation that demand dynamics differ meaningfully across locations (e.g., Kano vs. Lagos) and commodity types (e.g., vaccines vs. OTC drugs).

8.4 Tier 3 — Decomposition and Additive Models

Facebook Prophet (Meta, open-source) is the designated Tier 3 method. It was designed specifically for business and operational time series with the following characteristics that closely match this dataset:

- **Irregular seasonality:** Prophet decomposes series into yearly, weekly, and custom seasonal components, allowing it to capture annual procurement cycles (e.g., rainy-season disease burden, year-end budget spending) without manual specification.
- **Holiday and event effects:** Nigerian public holidays (Eid al-Fitr, Eid al-Adha, Christmas, Independence Day) and facility-specific events (donor supply arrivals, audit periods) can be encoded as regressors, preventing systematic forecast bias around these dates.
- **Missing data and outliers:** Prophet is robust to gaps in the time series — relevant here because some facility-month combinations have zero or missing procurement records due to stockouts or reporting lapses rather than true zero demand.
- **Uncertainty intervals:** Prophet natively produces credible intervals around each

forecast, which can be used directly to set reorder point buffers.

Prophet will be applied to category-facility pairs where Tier 2 models fail to achieve MAPE $\leq 20\%$, and as a parallel track to validate Tier 2 results on the highest-volume commodity categories.

8.5 Tier 4 — Machine Learning Models (Conditional)

Machine learning (ML) models will be evaluated **only if** the three tiers above do not meet the MAPE target across the majority of category-facility pairs. The primary candidates are:

- **Gradient Boosting (XGBoost / LightGBM):** Uses lag features (e.g., A_{t-1}, A_{t-3}, A_{t-12}), rolling statistics, and categorical covariates (facility, commodity category, quarter) as predictors. Can model cross-facility patterns jointly, which is an advantage when individual facility series are short. Requires careful feature engineering and time-respecting cross-validation to avoid data leakage.

- **Random Forest Regression:** Similar feature structure to gradient boosting but less prone to overfitting on small datasets. Serves as a lower-variance ML alternative.

> **Why deep learning is deferred:** Recurrent neural networks (LSTM, Temporal Fusion Transformers) require substantially more data per series than is currently available (36 months at category level). They also offer limited interpretability for operational decision-making. These approaches will be reconsidered at the SKU-level phase when data volume is higher.

8.6 Model Selection and Validation Strategy

Model selection will use **walk-forward (expanding-window) cross-validation** on the training data (2021–2023), followed by final evaluation on the held-out validation window (Jan–Jun 2024).

The procedure for each category-facility series is:

1. Train on Jan 2021 – Dec 2022; forecast Jan–Jun 2023; compute MAPE.

2. Extend training to Jun 2023; forecast Jul–Dec 2023; compute MAPE.

3. Average the two cross-validation MAPEs to rank candidate models.

4. Retrain the best-ranked model on the full training window (Jan 2021 – Dec 2023) and evaluate on the validation window (Jan–Jun 2024) to report the official Q2 baseline.

The winning model may differ by category-facility pair. A lookup table mapping each pair to its selected model will be maintained and updated as part of ongoing monitoring.

Tier	Models
Tier 1 – Benchmark	Seasonal Naïve, 12-Month Rolling Average
Tier 2 – Classical	ETS (Holt-Winters), SARIMA, Theta
Tier 3 – Additive	Facebook Prophet (with holiday regressors)
Tier 4 – ML (if needed)	XGBoost, LightGBM, Random Forest
Deferred	LSTM, Temporal Fusion Transformers

9. Scope Summary

Dimension	Decision
Commodities	7 health commodity categories (see Section 3)
Facilities	6 pilot facilities across Abuja, Kano, Lagos
Training data	January 2021 – December 2023
Validation data	January 2024 – June 2024
Forecast horizon	Monthly; 3-month rolling window
Granularity	Category × Facility × Month
Order filter	`state = done` only
Accuracy metric	MAPE per category-facility pair
Modelling tiers	Seasonal Naïve → ETS/SARIMA/Theta → Prophet → ML (escalate as needed)
Validation method	Walk-forward cross-validation; final eval on Jan–Jun 2024 hold-out

10. Open Questions for Stakeholder Alignment

The following questions require input from clinical and supply chain stakeholders before this scope is finalised:

1. **REACH Hub inclusion:** Are the REACH Hub clinics part of the formal pilot programme, or do they operate under a separate procurement protocol that would require a distinct forecasting track?

2. **Kano Distribution Warehouse:** Does the warehouse dataset reflect what was *dispatched to facilities* (downstream demand) or what was *purchased from suppliers* (upstream procurement)? The answer determines whether it can serve as a proxy for facility-level consumption.

3. **Critical commodity list:** Does the clinical or supply chain team maintain a formal list of "essential" or "critical" health commodities (analogous to the WHO Essential Medicines List or a national formulary)? If so, this list should anchor the in-scope product selection rather than category-level filtering alone.

11. Next Steps

Upon sign-off of this scope document, the following activities will commence:

1. **Data extraction and cleaning** — Filter the procurement dataset to the defined scope; handle missing values, zero-quantity records, and duplicate entries.

2. **Exploratory data analysis** — Characterise demand distributions, identify seasonality patterns, and flag facilities or categories with insufficient history.

3. **Baseline MAPE computation** — Apply a naïve benchmark model (e.g., prior-year same-month average) to the validation window and record the Q2 baseline MAPE per category-facility pair.

4. **Model training and selection** — Apply the four-tier modelling framework (Section 8) using walk-forward cross-validation; record the winning model per category-facility pair and report the official Q2 baseline MAPE.

5. **Monitoring pipeline** — Set up a repeatable monthly re-evaluation process to track MAPE improvement toward the Q3 target of $\leq 20\%$.